GOOGLE WORKSPACE TRAINING



# Google Sheets Foundation and Intermediate

## One-day course review draft

Build and check practical workplace outputs with fictional project data. These notes and prompts are a written review draft awaiting full live lab validation.



Delivered by Asraf · Microsoft Certified Trainer

**Google Sheets Fundamental and Intermediate**

One-day course review draft · Review draft, version 0.4

© 2026 Asraf Jaafar Sidik. All rights reserved.

Resources: Use the practice files supplied with this review package.

Google, Google Workspace, Gmail, Google Drive, Google Docs, Google Sheets, Google Slides, Google Sites, Gemini and NotebookLM are names associated with Google. This independent training guide is not published or endorsed by Google.

Screenshots show the product at the time of writing. Google updates its apps often, so labels, availability and layouts in your account may differ. This review draft still requires full lab validation.

# About the author

**Asraf** is a Microsoft Certified Trainer and HRD Corp-certified IT trainer based in Malaysia. He has been teaching working professionals since 2008, helping teams at banks, pharmaceutical companies, construction firms, and retailers get real work done with Microsoft 365, Power Platform, Excel, Google Workspace, and, more recently, AI tools like Copilot.

He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

# Contents


**Before you begin** 5

**1 Create and Navigate a Workbook** 6
1.1 Open a Native Copy 6
1.2 Navigate the Register 6
1.3 Manage a Practice Tab 7
1.4 Build a Workbook Map and Reconstruct a Small Register 7
Prompts and values 9

**2 Enter and Format Reliable Data** 10
2.1 Inspect Values and Types 10
2.2 Improve Readability 11
2.3 Inspect Formatting Effects 11
2.4 Diagnose Display, Type and Readability Problems 11
Prompts and values 13

**3 Calculate and Copy Formulas** 14
3.1 Calculate Variance 14
3.2 Summarise the Columns 15
3.3 Anchor an Assumption 15
3.4 Test Calculations With Zeros, Blanks and Reference Errors 15
Prompts and values 17

**4 Clean and Organise Records** 18
4.1 Clean a Small Import 18
4.2 Sort Whole Records 19
4.3 Freeze and Name a Range 19
4.4 Clean an Intake and Prove That Records Stay Together 19
Prompts and values 21

**5 Validate and Filter the Register** 22
5.1 Add a Status Dropdown 22
5.2 Create a Filter View 23
5.3 Test a Rule Safely 23
5.4 Design and Verify Entry Rules and Focused Views 23
Prompts and values 25

**6 Summarise and Visualise** 26
6.1 Count Statuses 26
6.2 Build a Useful Chart 27
6.3 Explain and Check 27
6.4 Build a Second Summary and Diagnose a Misleading Chart 27
Prompts and values 29

**7 Share and Prepare Output** 30
7.1 Inspect Access 30
7.2 Prepare the Print Range 30
7.3 Compare Output Options 31
7.4 Prepare a Readable, Scoped and Reproducible Output 31
Prompts and values 33

**8 Complete the Register Challenge** 34
8.1 Add a New Record 34
8.2 Repair and Reconcile 35
8.3 Explain the Result 35
8.4 Stress-Test the Extended Register and Complete the Handover 35
Prompts and values 37

INTRODUCTION

# Before you begin

This review book accompanies Google Sheets Fundamental and Intermediate. Work through its chapters in order and keep the original practice inputs unchanged. The same notes and prompts also appear in the local course website.

## Current Status

All chapters in this book are written drafts. Most have not yet been executed end to end in the application. The masterclass's reference brief chapter has a tested core path with a recorded recovery prompt; that does not validate the other exercises. Treat VERIFY notes and draft notices as production work still to complete.

## What You Need

Use a desktop browser and the Google account approved for the class. Confirm app and AI access beforehand. Edition, administrator controls and language can affect availability.

Download the supplied fictional practice pack. No real client or personnel record is needed. Dataset week fields are relative practice weeks, not an agreed pilot start date.

## How to Use the Exercises

Follow the numbered steps, attempt the output and then open the expected-result box. AI wording can vary; source facts and approval conditions must remain correct.

Values and prompts follow the notes. Replace source placeholders before submitting. Inspect proposed edits before applying them. Drafting does not authorise sending, inviting people or publishing.

## Course Timing

This course is planned for 1 day, with 390 teaching minutes per day, excluding breaks and lunch. Estimates include practice, checks and discussion and need a classroom pilot.

Each course supplies independent starting inputs. Use a stated recovery file if an earlier output is missing.

## Expanded Casework

Revision 0.4 adds casework, independent challenges and evidence gates to every chapter. The original short guided path alone is insufficient. Timings are targets, not measured learner results. Read each case input and keep baseline and scenario files separate.

01

CHAPTER 1

# Create and Navigate a Workbook

In this chapter you will create a native Sheet, rename tabs and enter a small set of records.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 40 minutes: 10 explanation/demonstration, 10 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 30 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** A named native workbook with an intact source and a separate practice tab.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Create and name a native Sheet.
- Navigate tabs, cells and ranges.
- Keep the input records intact.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Download TaskRegister.xlsx. This workbook starts at row 1 so cell references in the course match exactly.

## 1.1 Open a Native Copy

1. Create a Drive folder named `GW - Sheets Foundation - Practice` and upload TaskRegister.xlsx.
2. Open it in Sheets and use File, Save as Google Sheets if it is still in Excel format.
3. Name the native file `GW - Sheets - Task Register`.

> **What should you see?**
>
> The native workbook has Tasks and Summary tabs, with 12 task records.

## 1.2 Navigate the Register

1. Open Tasks. Click the name box and enter `A1:H13` to select the full source table.

2. Select D5 and read the owner from the formula bar. Identify the corresponding task ID in column A.
3. Open Summary and return to Tasks. Practise moving between the tabs without changing their data.

> **What should you see?**
>
> You can identify a cell address, a rectangular range and a worksheet tab.

## 1.3 Manage a Practice Tab

1. Duplicate Tasks and name the copy `Navigation Practice`.
2. On the duplicate, enter `Practice only` in J1 and resize the column.
3. Return to the original Tasks tab. Check that the source A1:H13 remains unchanged.

> **What should you see?**
>
> Experiments are on a clearly named copy; the baseline input is preserved.

## 1.4 Build a Workbook Map and Reconstruct a Small Register

1. Create a **Workbook Guide** tab. Write a purpose for Tasks, Summary and Navigation Practice, and identify which contains the maintained 12-record source.
2. Use the name box to visit A2, D5, F8 and H13 on Tasks. For each, record the cell address, column meaning, visible value and corresponding TaskID in the guide.
3. On a new **Entry Practice** tab type the eight source headers in A1:H1. Recreate the first three task records from Tasks using values, preserving each field's position.
4. Check each recreated row against the original by TaskID, not by appearance. Correct one copying error if found and record the check.
5. Practise selecting a cell, a whole row, a whole column and A1:H4. Record which selection would be appropriate for formatting a header versus moving a complete record.
6. Reorder Entry Practice next to Navigation Practice, then close and reopen the workbook from Drive. Use the guide to locate the original source.
7. Compare the original XLSX and the native spreadsheet by type, title and location. Add a note telling the next learner which one to edit.

> **What should you see?**
>
> Workbook Guide with four address-to-record checks. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Without using Recent, reopen the native file and locate the owner and actual cost for T008 using its record identity.
2. Create a two-row **New Entry Form** practice table with blank inputs and clear field labels. Do not append unfinished rows to Tasks.

3. Explain and demonstrate a cell address, a range, a tab and the source file identity. Have a partner use the guide or test it yourself after closing the workbook.

## Evidence to Hand Over

- Workbook Guide with four address-to-record checks.
- Recreated three-row entry table and a corrected/verified copy record.
- Independent source retrieval and a labelled new-entry practice area.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
|---|---|
| An address is mistaken for a permanent task identity | Use TaskID to identify a record because sorting can move it to another row. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

You can navigate and manage a native workbook. Next, make its data readable without changing its meaning.

**Check yourself:** What is the difference between a spreadsheet file and a worksheet tab?

# Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

## Part 1: Typed Values

```
GW - Sheets Foundation - Practice
```

```
GW - Sheets - Task Register
```

```
A1:H13
```

```
Navigation Practice
```

```
Practice only
```

## Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Beginner navigation brief: You inherit an XLSX, a native copy and several tabs. Another learner must be able to identify the maintained source and find T008’s owner and actual cost without guessing from row position.
```

02

CHAPTER 2

# Enter and Format Reliable Data

In this chapter you will use dates, numbers, currency, wrapping and sensible column widths.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** A readable register with numeric currency values and sensible dimensions.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Apply formats to typed numbers.
- Use wrapping and fitted dimensions.
- Distinguish appearance from the stored value.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use your original Tasks tab. If needed, reopen a fresh native copy of TaskRegister.xlsx.

## 2.1 Inspect Values and Types

1. Click G2 and inspect the formula bar. The value is numeric 600, not text containing RM.
2. Select G2:H13. Apply a currency format suitable for Malaysian ringgit, with no decimals for this exercise.
3. Select E2:E13 and use a whole-number format. DueWeek is a practice week index, not a calendar date.

**What should you see?**

Costs show their currency while remaining numeric. Week values still range from 1 to 4.

## 2.2 Improve Readability

1. Select A1:H1 and apply a consistent header style with readable contrast.
2. Widen Task and Owner so text is legible. Use text wrapping for the task descriptions if needed.
3. Fit row heights after wrapping. Keep each data column separate; do not merge cells inside the register.

**What should you see?**

Headers, names and cost figures are readable without cutting off text.

## 2.3 Inspect Formatting Effects

1. On Navigation Practice, enter a sample date in J2 and apply a date format; keep it outside the baseline register.
2. Compare a numeric value displayed as currency with a typed text value such as RM 600.
3. Undo the practice entries if they are no longer useful. Confirm that formatting the original costs did not change their values.

**What should you see?**

You can explain why number formatting does not convert arbitrary text into a number.

## 2.4 Diagnose Display, Type and Readability Problems

1. On **Entry Practice** create a **Type Tests** block outside the copied records. Enter numeric 600, text `RM 600`, a relative week 2 and a local practice date in separate labelled cells.
2. Inspect each formula-bar value and its display. Apply currency to the number and compare it with the text; formatting alone does not turn arbitrary text into a usable numeric input.
3. Add a temporary `=SUM(...)` check over the numeric and text cost cells. Record which value contributes and why the visually similar entries behave differently.
4. Use **Format**, **Number** to give costs a consistent RM display and DueWeek a whole-number display. Keep practice dates outside the source week column.
5. Apply wrapping, column widths and header contrast to the copied three-row table. Make each complete task description readable without merging source cells.
6. Use **Paste special**, **Values only** and **Format only** in separate practice cells. Compare the result to a normal paste and note which operation transfers each property.
7. Inspect the original Tasks numeric count and source values again. Formatting must not alter the twelve records or create a confirmed calendar date.

**What should you see?**

Type-test block with observed calculation behaviour. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Repair the card's three proposed entries in a separate intake table. Preserve raw input and record the clean value or clarification needed.
2. Prepare a readable three-record handout view with consistent currency and whole-week display. Prove the underlying numeric values remain numeric.
3. Give a partner two visually similar cost cells and ask them which one can be calculated reliably. Use formula-bar and SUM evidence in your explanation.

## Evidence to Hand Over

- Type-test block with observed calculation behaviour.
- Raw-to-clean intake table and readable entry-practice layout.
- Independent numeric evidence and unchanged 12-row source.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
| --- | --- |
| Currency looks correct but totals are wrong | Inspect numeric type and formula-bar value before changing more display formats. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

Typed values and readable formatting prepare the register for calculation.

**Check yourself:** Why should you avoid merging cells inside a dataset?

## Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

### Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Formatting intake: PlannedCost “RM 600” supplied as text; DueWeek “next Monday” with no agreed calendar start; a long Task description clipped in a narrow column. Treat type, missing date evidence and presentation as three different issues.
```

03

CHAPTER 3

# Calculate and Copy Formulas

In this chapter you will use arithmetic, SUM, AVERAGE and COUNT; explain relative and absolute references.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** Copied variance formulas, reconciled totals and a tested fixed reference.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Write arithmetic and aggregation formulas.
- Copy relative references correctly.
- Anchor a shared assumption.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks A1:H13. The workbook has no prefilled answer formulas.

## 3.1 Calculate Variance

1. In I1 type `Variance`. In I2 enter `=H2-G2`.
2. Fill I2 through I13 using the fill handle or copy and paste.
3. Inspect I2, I7 and I13 in the formula bar. Each formula should reference its own row. Negative variance means actual cost is below planned cost.

**What should you see?**

T001 has variance -50. The formula in I13 uses H13 and G13.

## 3.2 Summarise the Columns

1. In K1:L4 enter labels Planned Total, Actual Total and Variance Total in K2:K4.
2. Enter `=SUM(G2:G13)` in L2, `=SUM(H2:H13)` in L3 and `=SUM(I2:I13)` in L4.
3. In K6:L7 add Average Planned and Numeric Cost Count, using `=AVERAGE(G2:G13)` and `=COUNT(G2:G13)`.

**What should you see?**

Planned total is RM 10,400, actual total RM 6,800 and variance total -RM 3,600. There are 12 numeric planned costs.

## 3.3 Anchor an Assumption

1. On Navigation Practice, type `Practice multiplier` in K9 and 1.1 in L9. This is a hypothetical calculation, not a tax rate.
2. In J2 enter `=G2*$L$9` and fill through J13.
3. Inspect a later formula. Change L9 to 1.2, observe the results and restore 1.1.

**What should you see?**

The multiplier reference remains $L$9 while each row's G reference moves.

## 3.4 Test Calculations With Zeros, Blanks and Reference Errors

1. Create a **Calculation Tests** tab with Planned, Actual and Variance in A:C. Enter the four pairs on the card, leaving the designated blank empty.
2. Calculate Actual minus Planned for each complete numeric pair. Record separately why a blank Actual requires interpretation rather than automatically meaning confirmed zero expenditure.
3. Compare a real numeric zero with a blank and with text. Use COUNT and SUM in clearly labelled test cells to inspect numeric participation.
4. Create one deliberately wrong same-row calculation on the practice tab. Predict its result, inspect the formula bar and repair the reference.
5. Rebuild the practice multiplier calculation with both a relative and an absolute multiplier reference in two columns. Fill each down and compare the second and final rows.
6. Change the multiplier once, record the effect and restore it. Keep this hypothetical exercise separate from tax, policy or approved rates.
7. Reconcile SUM of row variances with total actual minus total planned in Tasks. If they differ, locate a specific row or range error rather than adjusting the total manually.

**What should you see?**

Four-case calculation test with a documented blank-information limitation. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Create an independent three-row estimate table with at least one zero actual cost and one positive variance. Calculate and explain the results without copying a finished answer column.
2. Shorten one total range on the practice table, then use a second calculation to detect and repair the omission.
3. Write a short explanation of negative variance, numeric zero, blank information and a fixed reference, each supported by an actual test cell.

## Evidence to Hand Over

- Four-case calculation test with a documented blank-information limitation.
- Reference-error and omitted-range diagnosis with repaired formulas.
- Reconciled baseline totals and independent estimate table.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
| --- | --- |
| A plausible total hides a copied-row error | Inspect row references and compare sum of variances against the difference of totals. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

You calculated and reconciled the register using visible source values and references.

**Check yourself:** Which part of a copied formula should move, and which should stay fixed?

# Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

## Part 1: Typed Values

```
Variance
```

```
=H2-G2
```

```
=SUM(G2:G13)
```

```
=SUM(H2:H13)
```

```
=SUM(I2:I13)
```

```
=AVERAGE(G2:G13)
```

```
=COUNT(G2:G13)
```

```
Practice multiplier
```

```
=G2*$L$9
```

## Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Calculation pairs Planned/Actual: 100/80; 100/120; 100/0; 100/blank. The final blank means
“not supplied” for this test, not a confirmed zero. Use this to distinguish calculation
mechanics from data meaning.
```

04

CHAPTER 4

# Clean and Organise Records

In this chapter you will find and replace, trim text, sort whole records and freeze the header row.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** A cleaned text example, whole-record sort and named cost range.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Clean text without altering source evidence.
- Sort whole records.
- Keep headers visible while scrolling.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks and download DirtyText.csv. Keep cleaning practice separate from the original register.

## 4.1 Clean a Small Import

1. Import DirtyText.csv into a new spreadsheet or separate practice tab. Check its three column headings.
2. In D1 type `CleanTeam`; in D2 enter `=TRIM(B2)` and fill to D4.
3. Compare B and D. Correct the lowercase status complete to the canonical Complete in a separate cleaned status column.

**What should you see?**

The trailing/leading spaces are removed in the cleaned output while original text remains inspectable.

## 4.2 Sort Whole Records

1. Return to Tasks and select A1:I13, including the variance column if you created it.
2. Use Data, Sort range, Advanced range sorting options. Mark that the range has a header row and sort by Team, then TaskID.
3. Locate T001 after sorting and check its owner, costs and variance still belong to that same record.

**What should you see?**

The task row stays intact. Sorting one column in isolation would break the record relationships.

## 4.3 Freeze and Name a Range

1. Use View, Freeze and freeze one row. Scroll down and verify the header stays visible.
2. Select G2:G13 and use the named-range control to name it `PlannedCosts`.
3. In a spare labelled cell enter `=SUM(PlannedCosts)` and compare it with the earlier planned total.

**What should you see?**

The named range sums to 10,400 and the frozen header remains visible.

## 4.4 Clean an Intake and Prove That Records Stay Together

1. Import or type the five-row cleaning intake on the copy-paste page into a new **Cleaning Intake** tab. Preserve a raw copy before editing.
2. Create CleanTeam, CleanStatus and Disposition columns. Use TRIM where appropriate, standardise obvious case differences and flag ambiguous statuses for clarification.
3. Identify the duplicate TaskID and compare its other fields before deciding whether it is a repeated identical record or a conflicting update. Do not delete a row solely because the name appears twice.
4. Use **Find and replace** restricted to the cleaned-status column for a known spelling repair. Preview the target scope and keep the raw source unchanged.
5. Sort the complete intake range by Team and TaskID. Before and after sorting compare three TaskID-to-owner or status relationships against the raw copy.
6. On a disposable copy, sort just one data column to demonstrate how relationships can break. Undo immediately and prove restoration with those three checks.
7. Create a named range for the cleaned numeric test costs if supplied, or use the existing PlannedCosts range. Inspect its exact bounds and compare its SUM with an explicit-range SUM.

**What should you see?**

Preserved raw intake, justified clean values and explicit exceptions. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Produce a clean intake and an exception list independently. Keep unresolved category and conflicting-duplicate decisions visible.
2. Add one new numeric row only to a named-range practice copy and inspect whether the named range includes it. Repair its bounds and compare the two total methods.
3. Hand over the raw copy, clean result and three row-identity checks so another person can verify the cleaning without trusting your description.

## Evidence to Hand Over

- Preserved raw intake, justified clean values and explicit exceptions.
- Whole-record sorting evidence and a restored isolated-column demonstration.
- Named-range boundary test and independently reconciled total.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
|---|---|
| A clean-looking row is associated with the wrong task | Compare complete TaskID relationships before and after the sort, not only totals. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

Cleaning, whole-record sorts and visible headers make the data safer to analyse.

**Check yourself:** What must stay together when sorting a register?

# Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

## Part 1: Typed Values

```
CleanTeam
```

```
=TRIM(B2)
```

```
PlannedCosts
```

```
=SUM(PlannedCosts)
```

## Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Cleaning intake (TaskID | Team | Status): C001 |  Planning  | complete; C002 | Procurement |
In Progress; C003 | Site Coordination | Blocked; C002 | Procurement | In Progress; C004 |
Planning | Not Started. C001 has surrounding team spaces. The duplicate is identical; Blocked
remains unresolved. These are separate practice records.
```

05

CHAPTER 5

# Validate and Filter the Register

In this chapter you will add a status dropdown, identify invalid entries and create a filter view.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** A status dropdown and two named filter views.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Restrict status values.
- Create and name a filter view.
- Check hidden rows before interpreting totals.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the original Tasks table, with all 12 baseline records still present.

## 5.1 Add a Status Dropdown

1. Select F2:F13. Open Data, Data validation and add a dropdown rule.
2. Use Complete, In Progress and Not Started as the three values. Set the rule to reject invalid data where that option is available.
3. Inspect one existing cell and confirm that its current status is retained.

**What should you see?**

The original values remain valid and future entries use the same vocabulary.

## 5.2 Create a Filter View

1. Select A1:I13 and open the filter-view option under Data. Create a new filter view.
2. Name it `Incomplete tasks` and use the Status filter to exclude Complete.
3. Count visible task IDs, then close the filter view and confirm that all 12 return.

**What should you see?**

Eight tasks are visible in the incomplete view. Closing it restores the full dataset.

## 5.3 Test a Rule Safely

1. Copy the validation rule to a spare test cell outside the register and try `Done`.
2. Check whether the rule rejects it or shows a warning, and discuss the difference.
3. Clear the spare test cell. Open your named filter view again to confirm it remains available.

**What should you see?**

You know the rule's actual behaviour and have not changed the baseline statuses.

## 5.4 Design and Verify Entry Rules and Focused Views

1. Create a **Rule Tests** tab with test value, expected accept/reject, observed result and repair columns. Plan tests before changing the rules.
2. Test the three canonical statuses and invalid **Done** using spare validated cells. Record rejection versus warning behaviour.
3. On an Entry Practice copy add DueWeek validation for whole practice weeks 1 through 4. Inspect the available custom-formula or number rule and test 0, 1, 4, 5 and 1.5.
4. Decide how blanks should be handled for a partially entered record. Test actual behaviour and document it; do not claim validation guarantees every required field is filled.
5. Create a second named filter view for Planning tasks not Complete. Count visible TaskIDs and compare them with a manual source scan.
6. Open and close each named view, confirming that all source rows return. Compare an ordinary SUM before and during filtering and record whether hidden rows still contribute.
7. Write a short entry guide explaining canonical statuses, relative weeks, the invalid-input message and what to do with an unclear category.

**What should you see?**

Expected-versus-observed rule matrix including week boundaries and blanks. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently build a small entry test with one valid row, one invalid status and one out-of-range week. Demonstrate the rules rather than merely listing them.
2. Create a named view answering “Which Procurement tasks are not complete?” and list its TaskIDs as evidence.
3. Check whether each new practice row lies inside the rule and view ranges. Repair a missing bound on the practice copy and retest.

## Evidence to Hand Over

- Expected-versus-observed rule matrix including week boundaries and blanks.
- Two useful named views with independently checked visible records.
- Entry guide and a repaired coverage/range test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
| --- | --- |
| A new row accepts invalid values | Inspect rule coverage as well as the dropdown definition; retest the actual new row. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

Validation prevents inconsistent entries and filter views support focused inspection.

**Check yourself:** Does hiding a row delete it or remove it from an ordinary SUM?

## Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

### Part 1: Typed Values

```
Incomplete tasks
```

```
Done
```

### Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Rule test values: status Complete, In Progress, Not Started, Done and blank; DueWeek 0, 1, 4, 5 and 1.5. DueWeek is a relative index for this training exercise, not an approved calendar schedule.
```

06

CHAPTER 6

# Summarise and Visualise

In this chapter you will build a simple conditional total and chart; ask Gemini to explain a checked formula if available.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 55 minutes: 10 explanation/demonstration, 25 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** A conditional summary and a source-linked status chart.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Calculate a conditional count and total.
- Choose and label a simple chart.
- Use AI to explain a formula without surrendering the check.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks and the starter Summary tab. Existing baseline status counts are four per category.

## 6.1 Count Statuses

1. In Summary B2 enter `=COUNTIF(Tasks!$F$2:$F$13,A2)` and fill through B4.
2. Add a total in B5 using `=SUM(B2:B4)` and label A5 Total.
3. In D1:E4 add team labels and use SUMIF over Tasks B2:B13 and G2:G13 to calculate planned cost by team.

**What should you see?**

Status counts are 4/4/4 and total 12. Planned team totals are Planning 2,800, Procurement 4,000 and Site Coordination 3,600.

## 6.2 Build a Useful Chart

1. Select Summary A1:B4, excluding the total row. Choose Insert, Chart.
2. Choose a column chart and title it `Task count by status`.
3. Inspect category labels and make sure no blank category or total has become a fourth status.

**What should you see?**

The chart shows three status categories with equal counts.

## 6.3 Explain and Check

1. If Gemini is available, paste your COUNTIF formula and ask it to explain the anchored range and relative criterion cell in the panel only.
2. Compare the explanation with the formula in B3.
3. Temporarily change T003 to Complete, observe 5/4/3 and restore it. Confirm the chart returns to baseline.

**What should you see?**

Both your explanation and the chart's update behaviour have been checked against an actual input change.

## 6.4 Build a Second Summary and Diagnose a Misleading Chart

1. Add a second chart for planned cost by team using the checked D:E summary. Label currency units and keep task counts separate from cost amounts.
2. Create a small **Summary Checks** block comparing total status counts with the number of TaskIDs and team planned totals with Tasks planned total.
3. On a chart copy deliberately include the status total row as a fourth category. Explain the double-counting risk, then repair the source range.
4. On a summary copy deliberately misspell one status label. Observe how COUNTIF and the chart respond. Repair the spelling rather than overwriting the result with a typed number.
5. Write two supported findings, one about task counts and one about planned cost distribution. Avoid claims about on-time success or realised savings.
6. Test a temporary change to one task's planned cost. Check the row, team summary, grand total and chart; then restore the source value.
7. Record each change and restoration in a check log with expected effect, observed effect and the range or label you inspected.

**What should you see?**

Two reconciled summaries/charts with distinct units. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Prepare an independent one-page dashboard on a new tab with two labelled charts, source links and two concise supported findings.
2. Diagnose the card’s suspicious chart and summary symptoms using the source cells. Repair the cause, not just the visual appearance.
3. Use a reviewer or fresh read-through to check units, population and total reconciliation, then record one correction or evidence for a no-change decision.

## Evidence to Hand Over

- Two reconciled summaries/charts with distinct units.
- Documented total-row and misspelt-category fault repairs.
- Independent dashboard, supported findings and restored source-change test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
| --- | --- |
| A missing count is hidden by typing a result | Repair the criterion or formula so later source changes still recalculate. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

A checked summary and chart communicate the register accurately.

**Check yourself:** Why should the total row normally be excluded from this chart's categories?

# Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

## Part 1: Typed Values

```
=COUNTIF(Tasks!$F$2:$F$13,A2)
```

```
=SUM(B2:B4)
```

```
Task count by status
```

## Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Troubleshooting card: A status chart shows Complete, In Progress, Not Started and Total.
Another summary says 4, 4 and 0 even though 12 task rows remain. Investigate source range and
exact criterion spelling before altering the chart design.
```

07

CHAPTER 7

# Share and Prepare Output

In this chapter you will inspect access, print a selected range and check scaling and page breaks.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 40 minutes: 10 explanation/demonstration, 10 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 30 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** Checked access and readable register and summary print previews.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Inspect sharing without accidental exposure.
- Prepare a readable print range.
- Distinguish printing from public publishing.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the checked register. No public publishing or email distribution is required.

## 7.1 Inspect Access

1. Open Share and inspect General access and the list of people.
2. Explain Viewer, Commenter and Editor roles using this fictional file.
3. Close without changing permissions unless the trainer assigns a named test recipient.

**What should you see?**

The file remains at its existing access level.

## 7.2 Prepare the Print Range

1. Select Tasks A1:I13. Open File, Print and choose Selected cells.

2. Inspect landscape orientation, fit-to-width scaling and whether the header can repeat.
3. Check every preview page for clipped labels or unreadably small text. Adjust layout rather than shrinking everything to fit one page.

**What should you see?**

The print preview includes the intended records and readable headers, without unrelated spare calculations.

## 7.3 Compare Output Options

1. Cancel printing. Inspect File, Share, Publish to web if it is available, then cancel without publishing.
2. Explain that publication can expose a selected sheet or spreadsheet to a broader audience depending on settings.
3. Describe when a PDF snapshot would be more appropriate than an editable or published file.

**What should you see?**

You can distinguish file access, a print/export copy and a published web view.

## 7.4 Prepare a Readable, Scoped and Reproducible Output

1. Write an output brief for a contributor who needs working input cells and a sponsor who needs a fixed summary. Identify the intended file and scope for each.
2. Inspect **Share** settings and current inheritance without changing access. Record proposed reader/editor roles separately from actual tested permissions.
3. Select the summary range and choose **File**, **Print**. Compare selected cells, current sheet and workbook scope where offered.
4. Choose orientation and scaling that preserve readable labels. Test repeated frozen rows and page breaks for a multi-page Tasks output when available.
5. Inspect every print-preview page, including the final page. Record whether helper cells, spare tests or unrelated tabs appear.
6. Save a PDF with a clear revision name. Open it and compare count/cost totals and row labels with the maintained source.
7. Change one heading in the working source and demonstrate that the old PDF remains unchanged. Produce a newly named export and update an output manifest.

**What should you see?**

Inspected access settings and proposed-role plan. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Prepare two independent outputs: a concise sponsor summary and a complete task listing with readable headings. Do not force both into the same one-page layout.
2. Have a reviewer use the manifest to identify the current source and current fixed copy. Repair ambiguous filenames or scope labels.
3. Explain which ranges should be edited, which calculated cells should be reviewed and why a fixed PDF is not a live tracker.

## Evidence to Hand Over

- Inspected access settings and proposed-role plan.
- Two scope-checked readable PDFs with the correct final-page content.
- Source-to-export comparison and a revision manifest that identifies the current copy.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
| --- | --- |
| The PDF contains practice/helper content | Inspect selected-cells versus current-sheet/workbook scope and regenerate the intended output. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

Checked access and readable output complete the register handover.

**Check yourself:** Which output would automatically reflect later source changes, and which is a snapshot?

# Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

## Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Output brief: Sponsor wants a readable summary of the 12-task baseline and planned costs.
Contributor needs all task rows and labels. Exclude Entry Practice, temporary rule tests and
unrelated helper notes from the sponsor handout.
```

08

CHAPTER 8

# Complete the Register Challenge

In this chapter you will add records, repair a formula and reconcile the totals independently.

> **Prompts and values:** use the copy-paste page for the exact values used below.

**Target allocation:** 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

**Your result:** A separate 13-record challenge copy with reconciled totals and chart.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

## What You Will Learn

- Extend a register deliberately.
- Repair a formula and reconcile totals.
- Explain a checked chart.

## Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Make a separate challenge copy of your workbook. The original 12-record baseline must stay intact.

## 8.1 Add a New Record

1. Add T013 for Planning, task `Check the handover`, owner Aina Rahman, DueWeek 4, Status Not Started, PlannedCost 200 and ActualCost 0.
2. Extend validation and copy the variance formula into its row.
3. Identify every bounded count, sum and named range that must include the new row, then extend them.

**What should you see?**

The challenge copy has 13 unique records and includes the new task in its analysis.

## 8.2 Repair and Reconcile

1. Deliberately change one copied variance reference in the challenge copy and predict its effect.
2. Repair it and reconcile planned total 10,600, actual total 6,800 and variance -3,800.
3. Check status counts Complete 4, In Progress 4 and Not Started 5.

**What should you see?**

The extended formulas, summary and chart all include the thirteenth task.

## 8.3 Explain the Result

1. Show a partner the new record and its contribution to the totals.
2. Explain a relative reference, a fixed reference and a validation rule in your workbook.
3. Leave the challenge copy clearly named and the original source baseline unchanged.

**What should you see?**

Your explanation is supported by cells and the maintained baseline is preserved.

## 8.4 Stress-Test the Extended Register and Complete the Handover

1. On the existing 13-record challenge copy, add a **Change Log** naming every range extended for T013: row formula, totals, named range, validation, filter views and summary.
2. Inspect the formula in every variance row and confirm the copied calculation follows its own record after any sort.
3. Create a final control table with record count, planned total, actual total, variance total and status-count sum. Reconcile those by at least two methods.
4. Use the final correction card to change T013 ActualCost from 0 to 150 only on the challenge copy. Predict and check its effects before inspecting the refreshed charts.
5. Prepare a focused view for incomplete Planning tasks and confirm that its record list includes T013.
6. Regenerate the sponsor output from the changed challenge state. Inspect its labels and final values and distinguish it from the untouched baseline export.
7. Write a short operating guide for entering the next record, handling invalid input, checking ranges and exporting a current summary.

**What should you see?**

13-record cost-corrected challenge and a full coverage/change log. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

## Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Close the earlier click instructions. Add T014 on a further disposable copy using the supplied fields, then extend and verify all affected features independently.

2. Use your control table to catch at least one deliberately shortened range. Correct it and explain the cause with cell evidence.
3. Give a partner the operating guide and ask them to find the current source and identify the checks for the next row. Record a handover improvement and leave all scenario copies clearly named.

## Evidence to Hand Over

- 13-record cost-corrected challenge and a full coverage/change log.
- Separate 14-record independent test with reconciled totals and repaired range omission.
- Readable current export plus a tested operating guide and preserved baseline.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

## Troubleshooting

| Symptom | What to check |
|---|---|
| The challenge contaminates the next baseline lesson | Keep named scenario copies and compare their population and totals before proceeding. |
| An input or earlier result is missing | Reopen the input named in Before You Begin; do not substitute private company data |
| A menu or feature is unavailable | Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback |
| The result differs from the checkpoint | Compare the exact source range, wording and conditions before repeating the action |

## Lesson Summary

You have built, checked and extended a usable register.

**Check yourself:** Which ranges needed extending after T013 was added?

# Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

## Part 1: Typed Values

```
Check the handover
```

## Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Corrections and extension: T013 ActualCost becomes 150. Further-copy T014: Procurement |
Verify submission checklist | Bala Kumar | week 4 | In Progress | planned 300 | actual 100.
Keep all three states separate: original baseline, 13-record corrected challenge, 14-record
extension.
```

# Google Sheets Fundamental and Intermediate

A practical 1-day course with step-by-step draft labs, copy-ready prompts, fictional inputs and expected-result checks. Use the supplied practice pack. Full lab execution, screenshots and classroom timing validation remain required before participant release.

Asraf Jaafar Sidik · Microsoft Certified Trainer ·